

Linear Interpolation

The process of making an assumption based off of data.

Piecewise linear

Each interval is described with a straight line

Consider $y_i : x_1 \leq x_i \leq x_2$

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1} = m$$

$\therefore$ The equation of a straight line going through (x_1, y_1) and

$(x_2, y_2) :$

$$y_i = \frac{(x_i - x_1)y_2 + (x_2 - x_i)y_1}{(x_2 - x_1)}$$

Investigating convergence of model

Does the model get better with more data?

consider $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Then $f(x+h) \approx f(x) + f'(x)h$

$$\lim_{n \rightarrow \infty} \frac{d^r y}{dx^n} (x-a)^r \sum_{n=0}^{\infty} \frac{f(x)}{n!}$$

$$r = \frac{1}{r!}$$